REVIEW FOR FINAL EXAM GRADE 10

A. Multiple choice questions

1. A runner is running around a circular track at a constant speed. What can you infer about his velocity?
a. Her velocity is non-constant **b.** Her velocity is constant
c. Her velocity is zero **d.** Her velocity doesn't exist

2. A body is thrown upwards without air resistance and reaches its maximum height. At that position:
a. Its velocity is zero and its acceleration is also zero.
b. Its velocity is zero but its acceleration is maximum.
c. Its acceleration is minimum.
d. Its velocity is zero and its acceleration is the acceleration due to gravity.

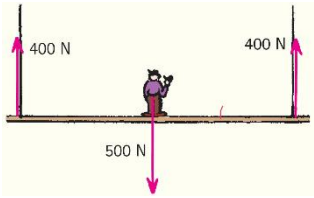
3. A boat moving away from you as you jump out of it demonstrates which of Newton's Laws?
a. Newton's 1st Law b. Newton's 2st Law
c. Newton's 3st Law d. None of these

4. A box (placed on a table) is being pushed by a force parallel to the table's surface. How many horizontal forces acting on the box are there?
a. Applied force and friction.
b. Gravitational force and normal force.
c. Applied force, friction, gravitational force, and normal force.
d. Applied force, friction, gravitational force, normal force, and thrust force.

5. An airplane slows down at a constant rate to land at an airport. What can you infer about its acceleration?
a. is non-constant (changing) **b.** is constant
c. is zero **d.** doesn't exist

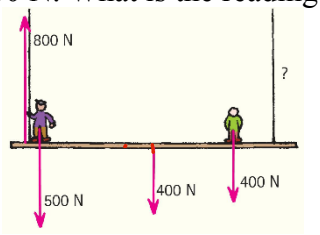
6. Two cars are moving along the same straight highway. Car A is moving northward at 90 km/h while car B is driving in the opposite direction at the same speed. What can you tell about their velocities?
a. $v_A = v_B = 90\,m/s$ b. $v_A = v_B = -90\,m/s$
c. $v_A = v_B$ d. $v_A = -v_B$

7. The sketch shows a painter's scaffold in the mechanical equilibrium. The person in the middle weighs 500 N, and the tension in each rope is 400 N. What is the weight of the scaffold?



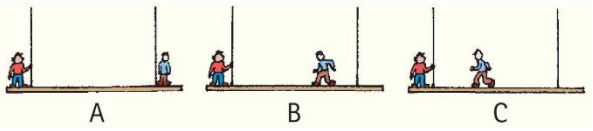
a. 100 N b. 300 N
c. 600 N d. 900 N

8. A different scaffold that weighs 400 N supports two painters, one 500 N and the other 400 N. The reading in the left scale in 800 N. What is the reading in the right-hand scale?



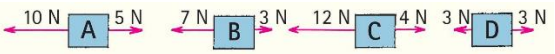
a. $500\,N$
b. $800\,N$
c. $1300\,N$
d. $400\,N$

9. The weights of Burl, Paul, and the scaffold produce tensions in the supporting ropes. Rank the tension in the left rope, from greatest to least, in the three situations A, B, and C.



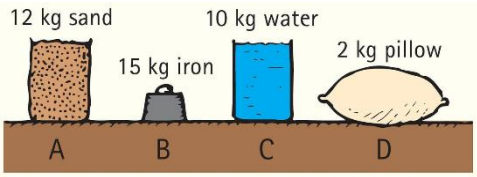
a. A, B, C
b. B, C, A
c. C, A, B
d. C, B, A

10. Rank the net force on the block from greatest to least in the four situations A, B, C, and D.



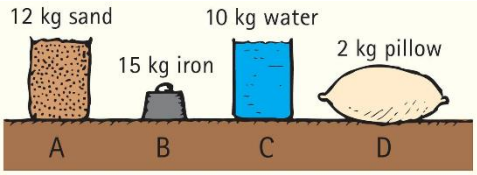
a. C, A, B, D
b. C, A, D, B
c. A, C, B, D
d. C, B, A, D

11. Different materials, A, B, C, and D, rest on a table.
a. From greatest to least, rank them by how much they resist being set in motion.



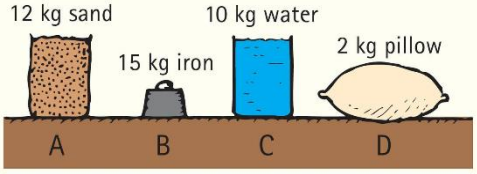
a. C, A, B, D
b. B, A, C, D
c. D, C, A, B
d. C, B, A, D

b. From greatest to least, rank them by the support (normal) force the table exerts on them.



a. B, A, C, D
b. C, A, B, D
c. D, C, A, B
d. C, B, A, D

c. Rank the objects inertia from greatest to least.

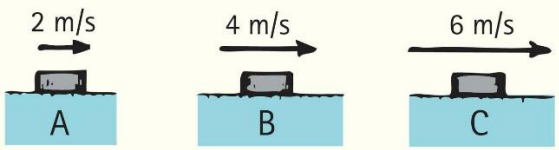


a. C, A, B, D
b. B, A, D, C
c. B, A, C, D
d. C, B, A, D

12. Three pucks, A, B, and C, are sliding across ice at the given speeds. Air and ice friction forces are negligible.

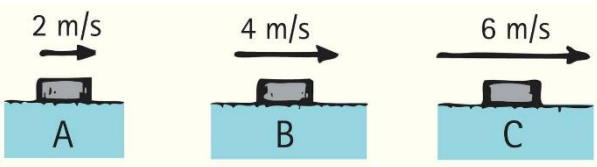
a. From greatest to least, rank them by the force needed to keep them moving.

a. C, A, B
b. C, B, A
c. $A = B = C$
d. B, C, A



b. From greatest to least, rank them by the force needed to stop them in the same time interval.

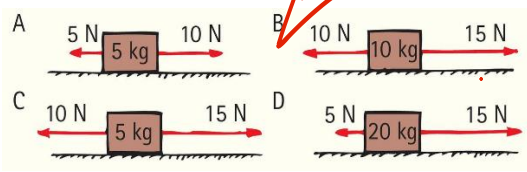
a. C, B, A
b. A, C, B
c. A, B, C
d. B, C, A



13. Boxes of various masses are on a friction-free level table. From greatest to least, rank the

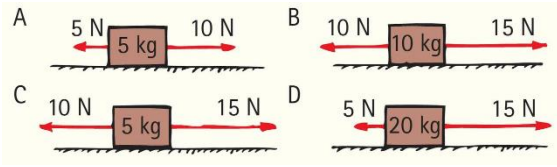
a. net forces on the boxes

a. A, B, C, D
b. D, C, B, A
c. $A = B = C = D$
d. B, C, A, D



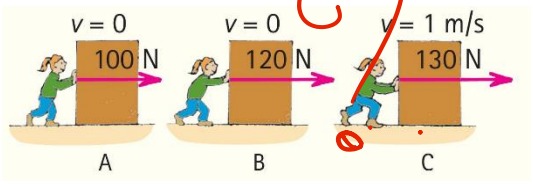
b. accelerations of the boxes.

a. C, A, B, D
b. $A = C, B, D$
c. $D, B, A = C$
d. $B, D, A = C$



14. In cases A, B, and C, the crate is in equilibrium (no acceleration). From greatest to least, rank the amount of friction between the crate and the floor.

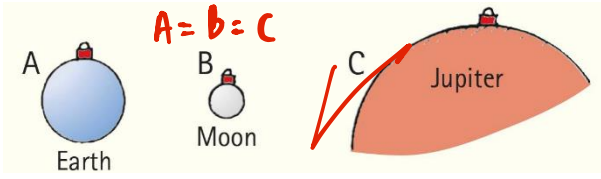
a. C, B, A
b. B, C, A
c. A, B, C
d. B, C, A



15. Consider a 100 kg box of tools in locations, A, B, and C. Rank from greatest to least the

a. masses of the 100 kg box of tools

a. C, B, A
b. A, C, B
c. B, A, C
d. C, A, B



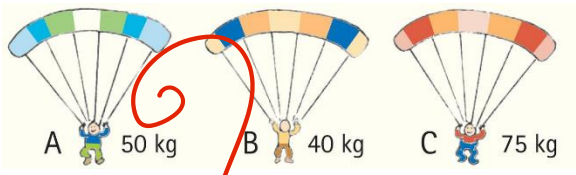
b. weights of the 100 kg box of tools.

a. B, A, C
b. A, B, C
c. C, A, B
d. C, B, A

16. Three parachutists, A, B, and C, each have reached terminal velocity at the same altitude.

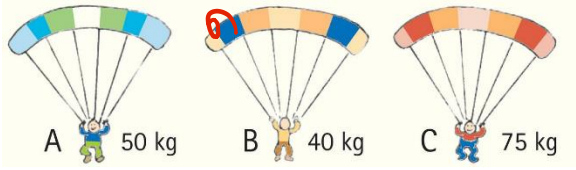
a. From fastest to slowest, rank their terminal velocities.

a. C, B, A
b. A, C, B
c. B, A, C
d. C, A, B



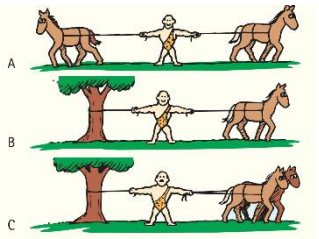
b. From longest to shortest times, rank their order in reaching the ground.

a. C, B, A
b. A, C, B
c. B, A, C
d. C, A, B



17. The strong man is pulled in the three situations shown. Rank the amount of tension in the rope in his right hand (the one attached to the tree in B and C) from least to greatest.

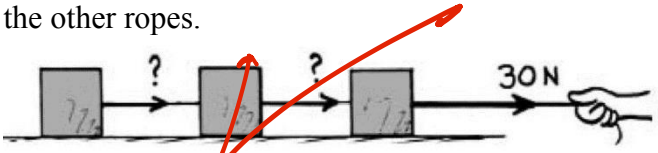
a. A, B, C
b. B, A, C
c. C, B, A
d. C, A, B



18. What quantity is represented by the area under a velocity-time graph?

a. Acceleration
b. Distance
c. Displacement
d. Speed

19. Three identical blocks are pulled, as shown, on a horizontal frictionless surface. If the tension in the rope help by the hand is 30 N, discuss what is the tension in the other ropes.



a. $T_1 = 10N, T_2 = 20N$
b. $T_1 = 20N, T_2 = 10N$
c. $T_1 = 30N, T_2 = 20N$
d. $T_1 = 20N, T_2 = 60N$

20. If a falling object gains $10\,m/s$ each second it falls, its acceleration is

a. $10\,m/s$
b. $10\,m/s$ per second
c. both of these
c. neither of these

21. Which of the following is a vector quantity?

a. Time taken
b. Energy
c. Acceleration
d. Temperature

22. A block of stone of mass $0.5\,kg$ is on the moon and it's being brought into the earth that has gravity greater than the moon. On the earth. the stone

a. has no mass and weight
b. has no mass and has weight
c. has mass and the same weight
d. has mass and weight heavier than.

23. According to 2nd law of Newton, which of the following statement is indeed correct?

a. Two objects have mass $m_1 > m_2$ exerted by the same force F, then the acceleration $a_1 > a_2$.

b. Two objects have mass $m_1 > m_2$ exerted by the same force F, then the acceleration $a_2 > a_1$.

c. Two objects have mass $m_1 > m_2$ exerted by the same force F, then the acceleration $a_1 = a_2$.

24. The equilibrium rule, $\Sigma F = 0$, applies to

a. objects or systems at rest.
b. objects or systems in uniform motion in a straight line.
c. both of these
d. neither of these

25. When a $10\,kg$ block is simultaneously pushed eastward with a force of $20\,N$ and westward with a force of $15\,N$, the combination of these forces on the block is

a. $35\,N$ west **b.** $35\,N$ east
c. $5\,N$ east **d.** $5\,N$ west

B. Critical thinking questions.

26. A car is driving in a circle with a constant speed of $30\,km/h$. Does the car maintain its velocity? Why or why not?

27. The force of air resistance acting on a certain falling object is roughly proportional to the square of the object's velocity and is directed upward. If the object falls fast enough, will the force of air resistance eventually exceed the weight of the object and cause the object to move upward explain.

C. Work problems

28. A race car travels on a racetrack at $44\,m/s$ and slows at a constant rate to a velocity of $24\,m/s$ over $11.0\,s$. How far does it move during this time?

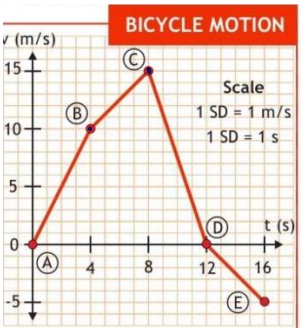
29. A woman driving at a speed of $20\,m/s$ sees a deer on the road ahead and applies the brakes when she is $800\,m$ from the deer. If the deer does not move and the car stops right before it hits the deer, what is the acceleration provided by the car's brakes?

30. The race car in the previous problems slows from $36\,m/s$ to $15\,m/s$ over $3.0\,s$. What is its average acceleration?

31. A car start moving from the rest acceleration of $1\,m/s^2$ about $200\,m$ from the starting point.

a. What is the car final velocity?
b. How long does the car take to reach that point?

32. The velocity time graph shows the motion of bicyclist in a straight line. (a) From the slope of the graph calculate the acceleration of the bicyclist between segment A and B, B and C, C and Dand D and E. (b) Calculate the average acceleration of the bicyclist. Also (C) Plot the acceleration time graph for this motion.



33. The force exerted by the wind on the sails of a sailboat is $390\,N$ north. The water exerts a force of $180\,N$ east. If the boat (including its crew) has a mass of $270\,kg$, what are the magnitude and direction of its acceleration?

34. A force F applied to an object of mass m_1, produces an acceleration of $3.00\,m/s^2$ on horizontal axis. The same force applied to a second object of mass m_2 produces an acceleration of $1.00\,m/s^2$ on horizontal axis.

a. What is the value of the ratio m_1/m_2 ?

b. If m_1 and m_2 are combined into one object, find its acceleration under the action of the force F.

35. Two objects are connected by a light string that passes over a frictionless pulley as shown in Figure P5.42. Assume the incline is frictionless and take $m_1 = 52.00\,kg$, $m_2 = 56.00\,kg$, and $\theta = 55.0^\circ$. a. Draw free-body diagrams of both objects. Find b. the magnitude of the acceleration of the objects, c. the tension in the string, and d. the speed of each object 2.00 s after it is released from rest.

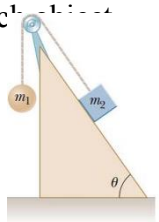
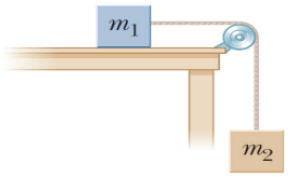


Figure P5.42

36. A race car can be slowed with a constant acceleration of $-11.0\,m/s^2$.

a. If the car is going $55\,m/s$, find the distance it travels before it stops.
b. Find the distance it takes to stop a car going twice as fast.

37. a. Show that $a = \left(\frac{m_2}{m_1 + m_2}\right) g$



b. If $m_1 = 5.00\,kg$ and $m_2 = 9.00\,kg$, calculate its acceleration?

38. Madison pushes with a 160-N horizontal force on a 20-kg crate of coffee resting on a warehouse floor. The force of friction on the crate is 80 N. What is Madison 'acceleration?

39. Leroy, who has a mass of 100 kg, is skateboarding at 9.0 m/s when he smacks into a brick wall and comes to a dead stop in 0.2 s.

a. Show that his deceleration is 45 m/s^2.

b. Show that the force of impact is 4500 N.

40. Sophia exerts a steady 40-N horizontal force on a 8-kg box resting on a lab bench. The box slides against a horizontal friction force of 24 N. What is the acceleration of the box.

41. An astronaut of mass 120 kg recedes from her spacecraft by activating a small propulsion unit attached to her back. The force generated by a spurt is 30 N. Show that her acceleration is 0.25 m/s^2.

42. A business jet of mass 30,000 kg takes off when the thrust for each of two engines is 30,000 N. What is its acceleration?

43. A rocket of mass 100,000 kg undergoes an acceleration of 2 m/s^2. What is the net force acting on it?

44. Allison exerts a steady net force of 50 N on a 20-kg shopping cart initially at rest for 2.0 s. Find the acceleration of the cart, and what is a distance moved by a cart?

45. The heavyweight boxing champion of the world punches a sheet of paper in midair, bringing it from rest up to a speed of 25.0 m/s in 0.050 s. The mass of the paper is 0.003 kg. Show that the force of the punch on the paper is only 1.50 N.

46. If you stand next to a wall on a frictionless skateboard and push the wall with a force of 40 N, how hard does the wall push on you? If your mass is 80 kg, what is your acceleration?

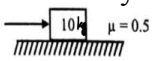
47. A force F acts in the forward direction on a cart of mass m. A friction force f opposes this motion.

a. Use Newton's second law and show that the acceleration of the cart is $(F - f/m)$.

b. If the cart's mass is 4.0 kg, the applied force is 12.0 N, and the friction force is 6.0 N, what is the cart's acceleration?

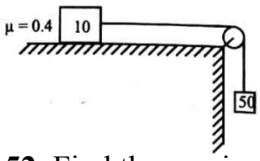
48. A rock band's tour bus, mass M, is accelerating away from a stop sign at rate a when a piece of heavy metal, mass $M/6$, falls onto the top of the bus and remains there. **a.** Show that the bus' acceleration is now $6/7\,a$. **b.** If the initial acceleration of the bus is 1.2 m/s^2, show that when the bus carries the heavy metal with it, the acceleration will be 1.0 m/s^2.

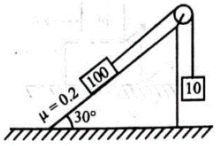
49. Find the acceleration and friction force for the body as shown in figure $(g = 10\,m/s^2)$



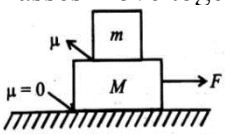
50. When a force of 50 N is applied on a body of mas 10 kg it is about to move and once the body is set into motion the same force of 50 N produces an acceleration of 0.5 m/s^2. Find coefficient of static and kinetic friction.

51. Find the acceleration of masses and tension in the string for the system shown $(g = 10\,m/s^2)$.

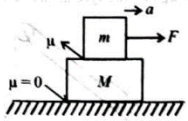




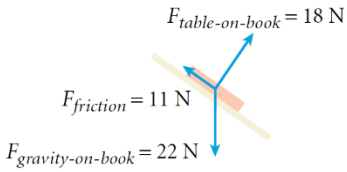
52. Find the maximum force applied on the lower mass so that the two masses move together, for the system of masses shown.



53. If the force is applied on the upper mass, then what will be your answer



54. Derek leaves his physics book on top of a drafting table that is inclined at a 35° angle. The free-body diagram at right shows the forces acting on the book. Find the net force acting on the book.



55. A 24 kg crate initially at rest on a horizontal floor requires a 75 N horizontal force to set it in motion. Find the coefficient of static friction between the crate and the floor.

56. A student attaches a rope to a 20.0 kg box of books. He pulls with a force of 90.0 N at an angle of 30.0° with the horizontal. The coefficient of kinetic friction between the box and the sidewalk is 0.500. Find the acceleration of the box.



26. No, the car does not maintain its velocity. While the car maintains a constant speed of 30 km/h, velocity is a vector quantity that includes both magnitude (speed) and direction. In circular motion, the direction of the car is constantly changing as it moves along the circular path. Therefore, the car's velocity is changing even though its speed remains constant.

27. Yes, if the object falls fast enough, the force of air resistance can eventually exceed the weight of the object and cause it to move upward. This is because the force of air resistance acting on an object is influenced by its velocity, and as the velocity of the object increases, the force of air resistance also increases.

28. Calculate the object moved during this time

Formula: $d = v_0 t + \frac{1}{2} a t^2$

since, $a = \frac{v - v_0}{t} = \frac{24\,m/s - 44\,m/s}{11.0\,s} = -1.82\,m/s^2$

So, $d = (44\,m/s)(11.0s) + \frac{1}{2}(-1.82\,m/s^2)(11.0s)^2$

$d = 484\,m - 110.11\,m = 373.89\,m$

29. Calculate the acceleration provided by a brake

Formula: $v^2 - v_0^2 = 2ad \Rightarrow a = \frac{v^2 - v_0^2}{2d}$

since, $v = 0\,m/s$ (stop), $v_0 = 20\,m/s$, $d = 800\,m$

So, $a = \frac{0^2 - (20\,m/s)^2}{800\,m} = -0.5\,m/s^2$

30. Calculate the average acceleration

Formula $a = \frac{\Delta v}{\Delta t} = \frac{v - v_0}{t_2 - t_1}$

since, $v = 15\,m/s$, $v_0 = 36\,m/s$, $\Delta t = t_2 - t_1 = 3s$

So, $a = \frac{15\,m/s - 36\,m/s}{3s} = -7\,m/s^2$

31. a. Calculate final velocity

Formula: $v^2 - v_0^2 = 2ad \Rightarrow v = \sqrt{v_0^2 + 2ad}$

since, $v_0 = 0\,m/s$ (rest), $a = 1\,m/s^2$, $d = 200\,m$

So, $v = \sqrt{0^2 + 2(1\,m/s^2)(200\,m)} = 20\,m/s$

b. Determine time taken to take a car to reach that point

Formula: $V = V_0 + at \Rightarrow t = \frac{V - V_0}{a}$

since, $V = 20 m/s$, $V_0 = 0$ (rest), $a = 1 m/s^2$

So, $t = \frac{20 m/s - 0 m/s}{1 m/s^2} = 20 s$

32a. Calculate the acceleration of the bicyclist between segment A & B

+ Segment A & B: Formula: $a = \frac{\Delta V}{\Delta t} = \frac{V_B - V_A}{t_B - t_A}$

since, $V_B = 10 m/s$, $V_A = 0 m/s$, $t_B = 4s$, $t_A = 0s$

So, $a = \frac{10 m/s - 0 m/s}{4s - 0s} = 2.5 m/s^2$

+ Segment B & C: Formula: $a = \frac{\Delta V}{\Delta t} = \frac{V_C - V_B}{t_C - t_B}$

since, $V_C = 15 m/s$, $V_B = 10 m/s$, $t_C = 8s$, $t_B = 4s$

So, $a = \frac{15 m/s - 10 m/s}{8s - 4s} = 1.25 m/s^2$

+ Segment C & D: Formula: $a = \frac{\Delta V}{\Delta t} = \frac{V_D - V_C}{t_D - t_C}$

since, $V_D = 0 m/s$ (origin), $V_C = 15 m/s$, $t_D = 12s$, $t_C = 8s$

So, $a = \frac{0 m/s - 15 m/s}{12s - 8s} = -3.5 m/s^2$

+ Segment D & E: Formula: $a = \frac{\Delta V}{\Delta t} = \frac{V_E - V_D}{t_E - t_D}$

since, $V_E = -5 m/s$, $V_D = 0 m/s$, $t_E = 16s$, $t_D = 12s$

So, $a = \frac{-5 m/s - 0 m/s}{16s - 12s} = -1.25 m/s^2$

b. Calculate the average acceleration of the bicyclist

+ Segment A & B: Formula: $d_{AB} = V_0 t + \frac{1}{2} a t^2$

since, $V_0 = V_A = 0 m/s$, $t = t_{AB} = 4s$, $a = 2.5 m/s^2$

So, $d_{AB} = (0 m/s)(4s) + \frac{1}{2}(2.5 m/s^2)(4s)^2 = 20 m$

+ Segment B & C: Formula: $d_{BC} = V_0 t + \frac{1}{2} a t^2$

since, $V_0 = V_B = 10 m/s$, $t = t_{BC} = 4s$, $a = 1.25 m/s^2$

So, $d_{BC} = (10 m/s)(4s) + \frac{1}{2}(1.25 m/s^2)(4s)^2 = 50 m$

+ Segment C & D Formula: $d_{CD} = V_0 t + \frac{1}{2} a t^2$

since, $v_0 = v_c = 15\,m/s$, $t = t_{CD} = 4s$, $a = -3.5\,m/s^2$

So, $d_{CD} = (15\,m/s)(4s) + \frac{1}{2}(-3.5\,m/s^2)(4s)^2$

$= 60m - 28m = 32m$

+ Segment D & E: Formula: $d_{DE} = v_0 t + \frac{1}{2}at^2$

since, $v_0 = v_D = 0\,m/s$, $t = t_{DE} = 4s$, $a = -1.25\,m/s^2$

So, $d_{DE} = (0\,m/s)(4s) + \frac{1}{2}(-1.25\,m/s^2)(4s)^2 = -10m$

Therefore, the total distance of the bicyclist is

$d = d_{AB} + d_{BC} + d_{CD} + d_{DE}$

$= 20m + 50m + 32m - 10m = 92m$

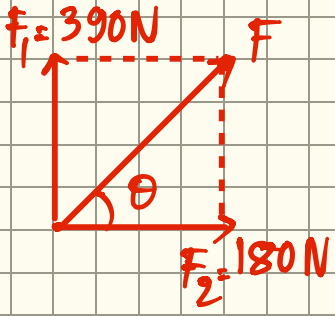
$t = 16s$

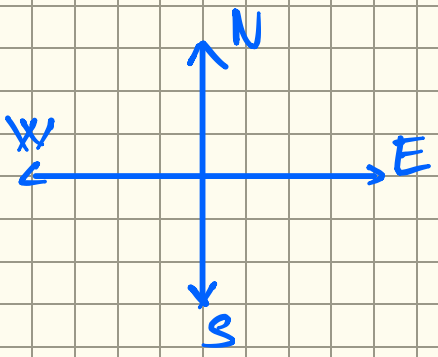
Formula: $v_{av} = \frac{d}{t} = \frac{92m}{16s} = 5.75\,m/s$

So, $v_{av} = 5.75\,m/s$

33. Calculate the magnitude and direction of its acceleration

Formula: $F = ma \Rightarrow a = \frac{F}{M}$





since, $m = 270kg$

$F = \sqrt{F_1^2 + F_2^2}$

$F = \sqrt{390^2 + 180^2} = 429.5N$

So, $a = \frac{429.5N}{270kg} = 1.59\,m/s^2$ in direction of $[N, E] = 65^\circ$

Or $\theta = \tan^{-1}\left(\frac{opp}{adj}\right) = \tan^{-1}\left(\frac{390}{180}\right) = 65^\circ$

34. a) Determine the value of the ratio m_1/m_2

- For mass m_1: $F = m_1 a_1$ (1)
- For mass m_2: $F = m_2 a_2$ (2)

(1) = (2) $\Rightarrow m_1 a_1 = m_2 a_2 \Rightarrow \frac{m_1}{m_2} = \frac{a_2}{a_1}$

since, $a_1 = 3.00\,m/s^2$, $a_2 = 1.00\,m/s^2$

So, $\frac{m_1}{m_2} = \frac{1.00\,m/s^2}{3.00\,m/s^2} = 1:3$

b). Find the acceleration in case m_1 & m_2 are combined into one object

Formula: $F = ma \Rightarrow a = \frac{F}{m}$

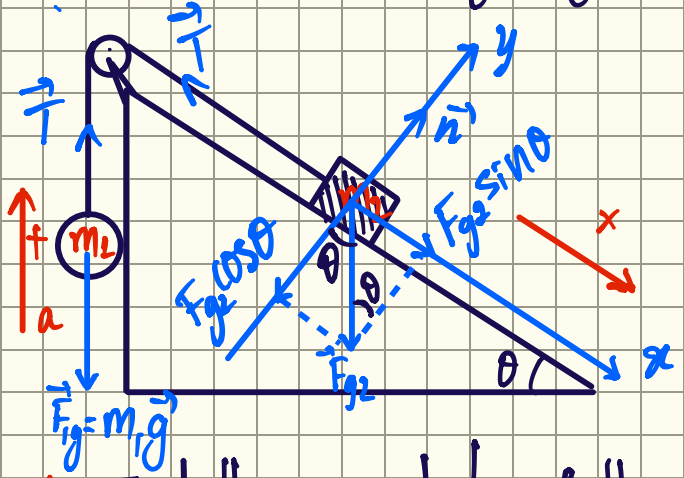
since, $m = m_1 + m_2$, $m_2 = 3m_1$

$m = m_1 + 3m_1 = 4m_1$

$F = m_1 a_1$

So, $a = \frac{m_1 a_1}{4m_1} = \frac{1}{4} a_1 = \frac{1}{4} \times 3\,m/s^2 = 0.75\,m/s^2$

35. a). Draw free-body diagrams of both object



b). Find the magnitude of the acceleration of the object

- For mass m_1: $\Sigma\vec{F} = m\vec{a} \Leftrightarrow \vec{F}_{g1} + \vec{T} = m_1\vec{a}$

$-F_{g1} + T = m_1 a \Leftrightarrow -m_1 g + T = m_1 a$

$\Rightarrow T = m_1 g + m_1 a$ (1)

- For mass m_2: $\Sigma\vec{F} = m\vec{a}$

$\vec{ox}$: $\vec{F}_{g2x} + \vec{T} = m_2\vec{a} \Leftrightarrow F_{g2x} - T = m_2 a$

$F_{g2}\sin\theta - T = m_2 a \rightarrow m_2 g \sin\theta - T = m_2 a$ (2)

$\vec{oy}$: $\vec{F}_{g2y} + \vec{n} = \vec{0} \Leftrightarrow F_{g2y} = n \rightarrow F_{g2}\cos\theta = n$

or $m_2 g\cos\theta = n$

(2) $\Rightarrow m_2 g\sin\theta - (m_1 g + m_1 a) = m_2 a$

$m_2 g\sin\theta - m_1 g - m_1 a = m_2 a$

$m_2 g\sin\theta - m_1 g = (m_2 + m_1)a$

$\Rightarrow a = \dfrac{m_2 g\sin\theta - m_1 g}{m_2 + m_1}$

since, $m_2 = 56\,kg$, $m_1 = 52\,kg$, $g = 10\,N/kg$, $\theta = 55°$

So, $a = \dfrac{(56\,kg)(10\,N/kg)\sin 55° - (52\,kg)(10\,N/kg)}{56\,kg + 52\,kg} = -0.57\ m/s^2$

Hence, $a = -0.57\,m/s^2$

c). Find the tension in the string

(1) We have $T = m_1 g + m_1 a$

$= (52\,kg)(10\,N/kg) + (52\,kg)(-0.57\ m/s^2)$

$= 490.36\,N$

So, $T = 490.36\,N$

d). Find the speed of each object 2.00s after it is released

Formula: $v = v_0 + at$

since, $a_1 = a_2 \Rightarrow v_1 = v_2$

So, $v_1 = v_2 = v_0 + at = 0 + (-0.57\,m/s^2)(2s)$

$$\boxed{v_1 = v_2 = -1.14\ m/s}$$

36. Find the distance it travels before it stops

Formula: $v^2 - v_0^2 = 2ad \Rightarrow d = \dfrac{v^2 - v_0^2}{2a}$

since, $v = 0\,m/s$, $v_0 = 55\,m/s$, $a = -11.0\,m/s^2$

So, $d = \dfrac{(0\,m/s)^2 - (55\,m/s)^2}{2(-11.0\,m/s^2)} = 137.5\,m$

b. Find the distance it takes to stop a car going twice as fast

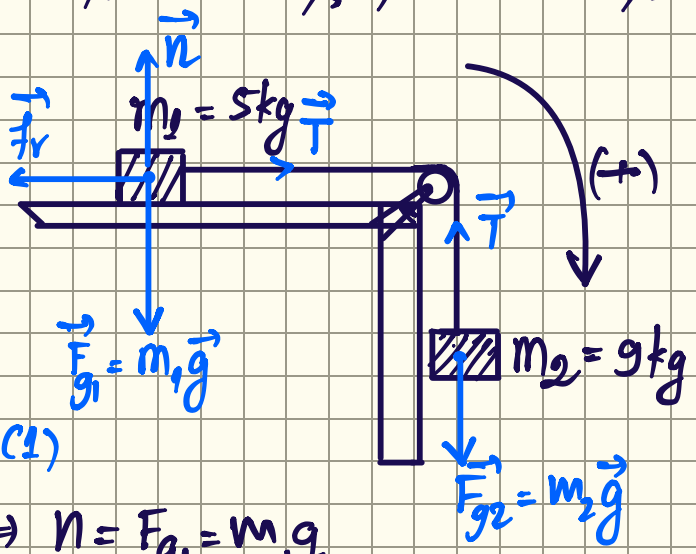
Formula $v^2 - v_0^2 = 2ad \Rightarrow d = \dfrac{v^2 - v_0^2}{2a}$

since, $v = 0$ (stop), $v = 2 \times 55\,m/s = 110\,m/s$, $a = -11.0\,m/s^2$

So, $d = \dfrac{(0\,m/s)^2 - (110\,m/s)^2}{-11.0\,m/s^2} = 220\,m$

37. a. Show that $a = \left(\dfrac{m_2}{m_1 + m_2}\right) g$

Formula $\Sigma \vec{F} = m\vec{a}$



- For mass m_1

$\vec{ox}: \vec{T} + \vec{f_r} = m\vec{a} \Leftrightarrow T - f_r = m_1 a$ (1)

$\vec{oy}: \vec{n} + \vec{F}_{g1} = \vec{0} \Leftrightarrow n - F_{g1} = 0 \Leftrightarrow n = F_{g1} = m_1 g$

since, $f_r = \mu n = \mu m_1 g$

(1) $\Rightarrow T - \mu m_1 g = m_1 a \Rightarrow T = m_1 a + \mu m_1 g$

- For mass m_2

$(\vec{oy}): \vec{F}_{g2} + \vec{T} = m_2\vec{a} \Leftrightarrow F_{g2} - T = m_2 a \Leftrightarrow m_2 g - T = m_2 a$ (2)

Sub (1) → (2): $m_2 g - (m_1 a + \mu m_1 g) = m_2 a$

$m_2 g - m_1 a - \mu m_1 g = m_2 a$ (3)

Assume that, there is no friction, so that the term $\mu m_1 g$ is neglected.

(3) $\Rightarrow m_2 g - m_1 a = m_2 a \Leftrightarrow m_2 g = m_1 a + m_2 a$

$m_2 g = (m_1 + m_2) a \Rightarrow a = \left(\dfrac{m_2}{m_1 + m_2}\right) g$

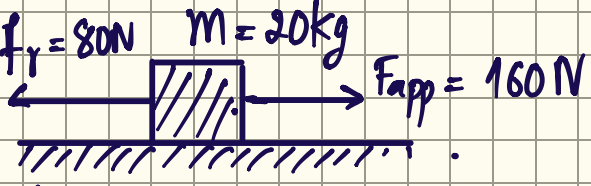
So, $\boxed{a = \left(\dfrac{m_2}{m_1 + m_2}\right) g}$

b). Calculate its acceleration if $m_1 = 5kg$, $m_2 = 9kg$

We have, $a = \left(\dfrac{m_2}{m_1 + m_2}\right) g$

So, $a = \left(\dfrac{9kg}{5kg + 9kg}\right)(10N/kg) = 6.43\,m/s^2$

38. Calculate acceleration of Madison



Formula: $\Sigma F = ma \Rightarrow a = \frac{\Sigma F}{m}$

since, $\Sigma F = 160N - 80N = 80N$, $m = 20kg$

So, $a = \frac{80N}{20kg} = 4 m/s^2$

39.a. Show that his acceleration is $45 m/s^2$

Formula $a = \frac{v - v_0}{t}$ since, $v_0 = 0$, $v = 9 m/s$, $t = 0.2s$

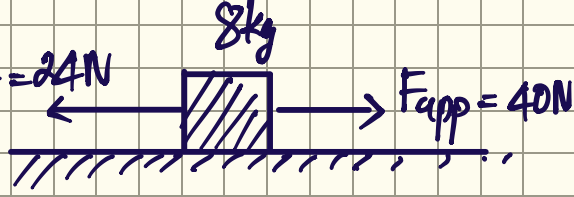
So, $a = \frac{9 m/s - 0 m/s}{0.2 s} = 45 m/s^2$

b. Show that the force of impact is 4500N

Formula $F = ma$, since, $m = 100kg$, $a = 45 m/s^2$

So, $F = (100kg)(45 m/s^2) = 4500N$

40. Calculate the acceleration of the box



Formula $\Sigma F = ma \Rightarrow a = \frac{\Sigma F}{m}$

since, $\Sigma F = F_{app} - f_r = 40N - 24N = 16N$, $m = 8kg$

So, $a = \frac{16N}{8kg} = 2 m/s^2$

41. Show that her acceleration is $0.25 m/s^2$

Formula $F = ma \Rightarrow a = \frac{F}{m}$

since, $F = 30N$, $m = 120kg$

So, $a = \frac{30N}{120kg} = 0.25 m/s^2$

42. Calculate its acceleration

Formula $F = ma \Rightarrow a = \frac{F}{m}$

since, $F = 30000N$, $m = 30000kg$

So, $a = \frac{30000N}{30000kg} = 1 m/s^2$

43. Calculate the net force

Formula $F = ma$, since, $m = 100000kg$, $a = 2 m/s^2$ So, $F = 200000N$

44. Calculate a distance moved by a cart

Formula $d = v_0 t + \frac{1}{2} a t^2$

since, $v_0 = 0$ (rest), $t = 2.0s$

$$a = \frac{F}{m} = \frac{50N}{20kg} = 2.5 m/s^2$$

So, $d = (0m/s)(2.0s) + \frac{1}{2}(2.5m/s^2)(2s)^2 = 5m$

45. Show that the force of the punch on the paper is only 1.50N

Formula $F = ma$

since, $m = 0.003kg$, $v_0 = 0$ (rest), $v = 25m/s$, $t = 0.050s$

$$a = \frac{v - v_0}{t} = \frac{25m/s - 0m/s}{0.050s} = 500 m/s^2$$

So, $F = (0.003kg)(500m/s^2) = 1.50N$

46. Calculate the acceleration

Formula $F = ma \Rightarrow a = \frac{F}{m}$

since, $F = 40N$, $m = 80kg$

So, $a = \frac{40N}{80kg} = 0.5 m/s^2$

47. a Show that the acceleration of the cart is $a = \frac{F - f}{m}$

Formula $\Sigma F = ma \Rightarrow a = \frac{\Sigma F}{m}$



since, $\Sigma F = F - f$, $m = m$

So, $a = \frac{F - f}{m}$

b. Calculate the cart's acceleration

We have $a = \frac{F - f}{m}$

since, $m = 4kg$, $F = 12.0N$

$f = 6.0N$

So, $a = \frac{12.0N - 6.0N}{4kg} = 1.5 m/s^2$

48 a. Show that the bus's acceleration is now $6/7\,a$

Formula: $F = ma$

- A rock band's tour bus of mass M: $F = Ma$ (1)
- Mass $M/6$ falls onto the top of the bus: $F = \left(M + \frac{M}{6}\right)a'$ (2)

since (1) = (2) $\Rightarrow$ $Ma = \left(M + \frac{M}{6}\right)a' \Rightarrow a' = \frac{Ma}{\left(M + \frac{M}{6}\right)} = \frac{Ma}{\frac{6M+M}{6}}$

$$a' = \frac{6Ma}{7M} = \frac{6}{7}a$$

So, $\boxed{a' = \frac{6}{7}a}$

b. Show that the acceleration $a' = 1\,m/s^2$

We have, $a' = \frac{6}{7}a$

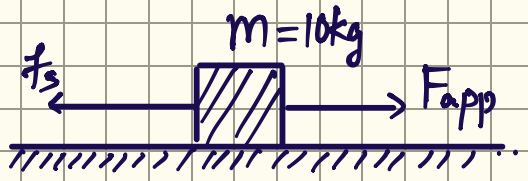
since, $a = 1.2\,m/s^2$

So, $a' = \frac{6}{7} \times 1.2\,m/s^2 = 1.03\,m/s^2 \simeq 1.0\,m/s^2$

49. Find the acceleration and friction force

Formula: $F = ma \Rightarrow a = \frac{F}{m}$ since, $F_{app} = 20N$, $m = 10\,kg$

So, $a = \frac{20N}{10kg} = 2\,m/s^2$



- Friction force

Formula: $f_s = \mu N = \mu mg$

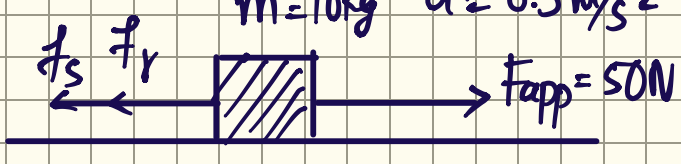
since, $\mu = 0.5$, $m = 10kg$, $g = 10N/kg$

So, $f_s = (0.5)(10kg)(10N/kg) = 50N$

50. Find coefficient of static and kinetic friction

Formula: $\Sigma F = ma$

since $\Sigma F = F_{app} - f_r = ma$



$\Rightarrow f_r = F_{app} - ma = 50N - (10\,kg)(0.5\,m/s^2) = 45N$

- Kinetic friction: $f_r = \mu_k n = \mu_k mg \Rightarrow \mu_k = \frac{f_r}{mg}$

So, $\mu_k = \frac{45N}{(10kg)(10N/kg)} = 0.45$

• Static friction: $f_s \geq F = ma$

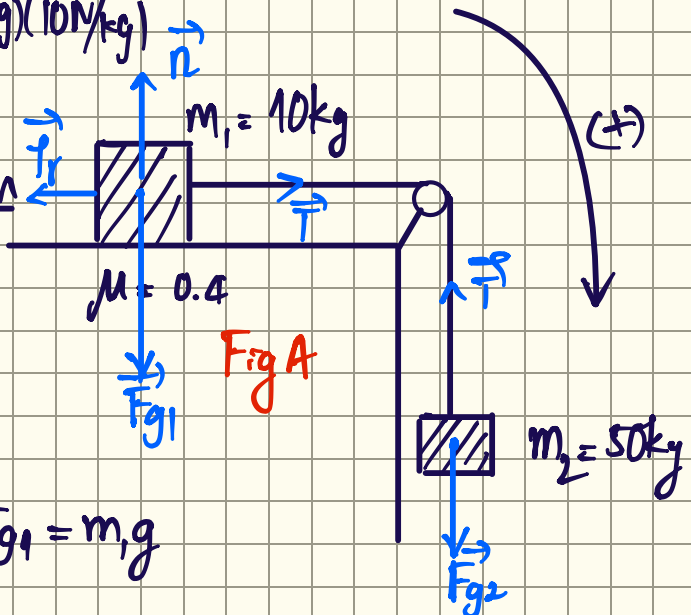
$\Sigma F = F_{pp} - f_s = ma \quad (a = 0)$

$F_{app} - f_s = 0 \Rightarrow f_s = F_{app} = 50N$

since, $f_s = \mu_s N = \mu_s mg \Rightarrow \mu_s = \frac{f_s}{mg} = \frac{50N}{(10kg)(10N/kg)} = 0.5$

So, $\mu_s = 0.5$

S1. Find the acceleration of masses and tension



Formula $\Sigma \vec{F} = m\vec{a}$

• For mass m_1

$\vec{ox}: \vec{T} + \vec{f_r} = m_1\vec{a} \Leftrightarrow T - f_r = m_1 a \quad (1)$

$\vec{oy}: \vec{n} + \vec{F_{g1}} = \vec{0} \Leftrightarrow n - F_{g1} = 0 \Rightarrow n = F_{g1} = m_1 g$

• friction force: $f_r = \mu n = \mu m_1 g$

$(1) \Rightarrow T - \mu m_1 g = m_1 a \Rightarrow T = m_1 a + \mu m_1 g$

• For mass m_2

$\vec{oy}): \vec{F_{g2}} + \vec{T} = m_2\vec{a} \Leftrightarrow F_{g2} - T = m_2 a \Rightarrow m_2 g - T = m_2 a \quad (2)$

Sub (T) into (2) $\Leftrightarrow$ $m_2 g - (m_1 a + \mu m_1 g) = m_2 a$

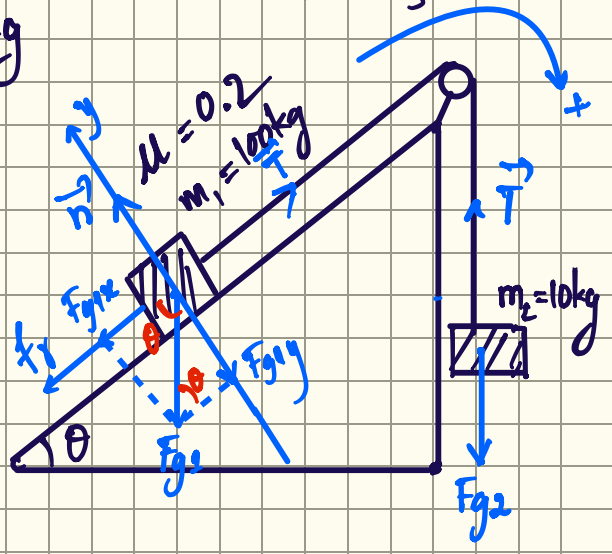
$m_2 g - m_1 a - \mu m_1 g = m_2 a$

$m_2 g - \mu m_1 g = (m_1 + m_2) a$

$\Rightarrow a = \frac{(m_2 - \mu m_1) g}{m_1 + m_2} = \frac{(50kg - 0.4 \times 10kg)\, 10N/kg}{10kg + 50kg} = 7.7 m/s^2$

So, $a = 7.7 m/s^2$

+ Find the acceleration and tension



• For mass m_1:

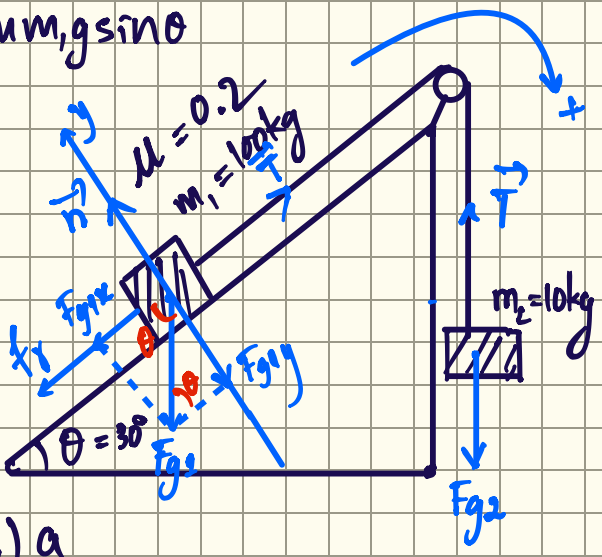
$\vec{ox}: \vec{T} + \vec{f_r} = m_1\vec{a} \Leftrightarrow T - f_r = m_1 a \quad (1)$

$\vec{oy}: \vec{n} + \vec{F_{g1y}} = \vec{0} \Leftrightarrow n - F_{g1y} = 0$

$n = F_{g1y} = m_1 g \sin\theta$

since, $f_r = \mu n = \mu m_1 g \sin\theta$

(1) $\Rightarrow T - \mu m_1 g \sin\theta = m_1 a \Rightarrow T = m_1 a + \mu m_1 g \sin\theta$



. For mass m_2

$(\vec{oy})$: $\vec{F_{g2}} + \vec{T} = m_2 \vec{a}$

$F_{g2} - T = m_2 a \Leftrightarrow m_2 g - T = m_2 a$ (2)

Sub (1) → (2): $m_2 g - (m_1 a + \mu m_1 g \sin\theta) = m_2 a$

$m_2 g - m_1 a - \mu m_1 g \sin\theta = m_2 a$

$m_2 g - \mu m_1 g \sin\theta = (m_1 + m_2) a$

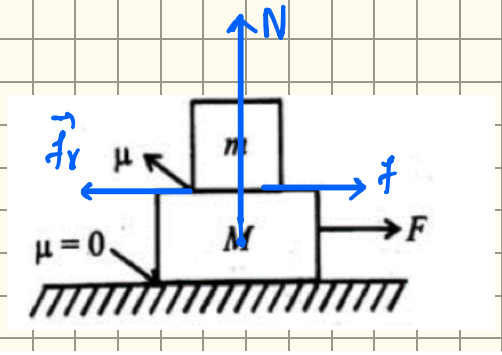
$$\Rightarrow a = \frac{m_2 g - \mu m_1 g \sin\theta}{m_1 + m_2} = \frac{(m_2 - \mu m_1 \sin\theta) g}{m_1 + m_2}$$

$$= \frac{(10kg - 0.2 \times 100kg \times \sin 30°)\, 10N/kg}{100kg + 10kg} = 0$$

So, $a = 0$

52 Find the maximum force on the lower mass

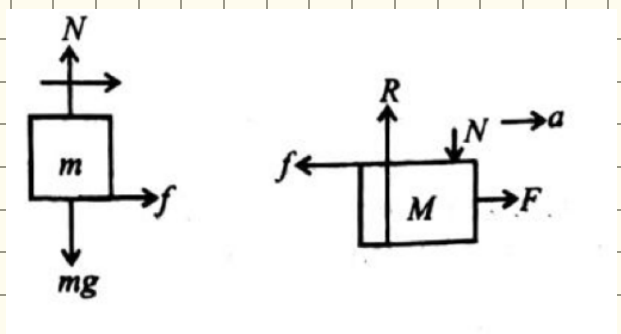
Formula $\Sigma F = ma$



. For mass M: $F - f = Ma$

since, $f = ma \Rightarrow F - ma = Ma$

$F = (m + M) a \Leftrightarrow a = \frac{F}{M+m}$



. For mass m: $F - f = Ma$

$F - \mu mg = Ma$

$F = \mu mg + Ma$

$= \mu mg + M\left(\frac{F}{M+m}\right)$

$F(M+m) = \mu mg(M+m) + MF$

$FM + Fm = \mu mg(M+m) + MF$

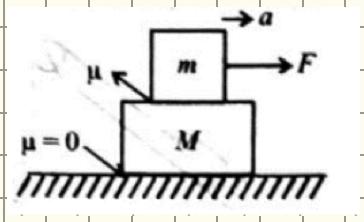
So, $\boxed{F = \mu g(M+m)}$

Also $F - f = Ma \Rightarrow F - ma = Ma \Rightarrow \boxed{a = \frac{F}{M+m}}$

$F = f + ma$

$F = \mu mg + M\frac{F}{M+m} \Rightarrow F - \frac{MF}{m+M} = \mu mg \Rightarrow \boxed{F = \mu g(M+m)}$

53. Calculate the force F



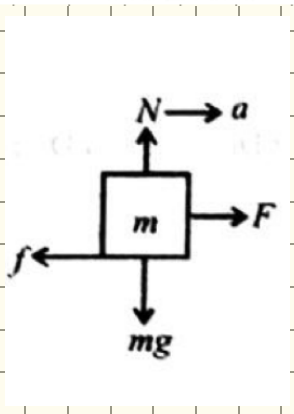
As the two masses are moving together, so $a = \frac{F}{M+m}$

friction force $f = \mu N = \mu mg$

since, $F - f = ma$

$F - \mu mg = ma$

$F - \mu mg = m\left(\frac{F}{M+m}\right)$

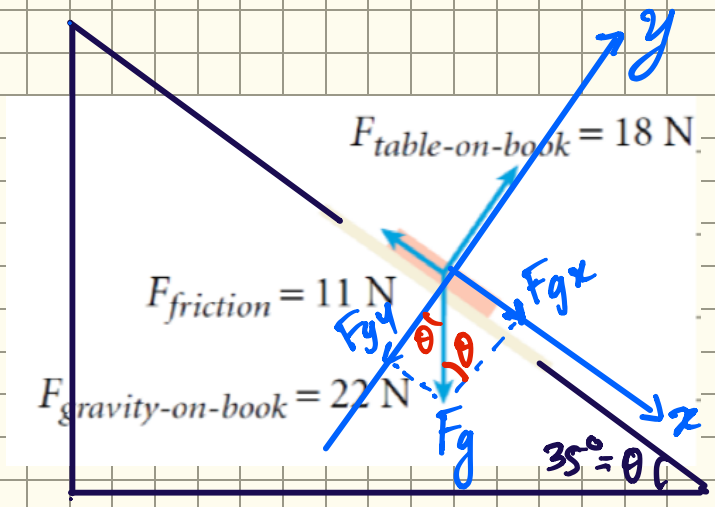


$F - \frac{mF}{M+m} = \mu mg \Leftrightarrow F\left(1 - \frac{m}{M+m}\right) = \mu mg$

$\Rightarrow F = \frac{\mu mg}{M}(M+m)$

So, $\boxed{F = \frac{\mu mg}{M}(M+m)}$

54. Find the net force acting on the book



The net force: $\vec{F}_{net} = \vec{F}_x + \vec{F}_y$

$F_{net} = \sqrt{F_x^2 + F_y^2}$

$\overrightarrow{ox}$: $\vec{F}_x$: $\vec{F}_{gx} + \vec{F}_r = 0$

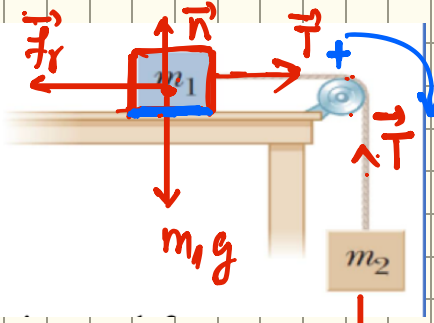
$F_{gx} - F_r = 0$

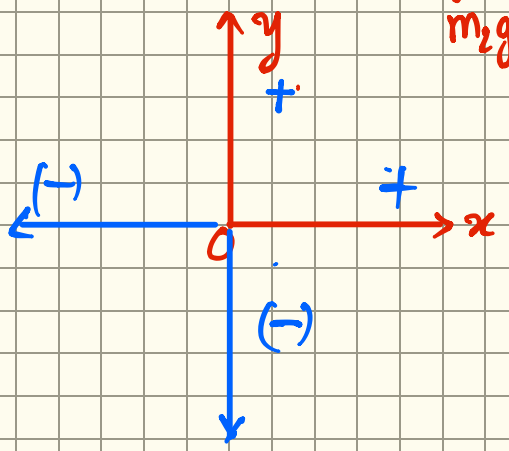
$F_g \sin\theta - F_r = 0$

$F_r = F_g \sin\theta = (22\text{N}) \sin 35^\circ$

37. a. Show that $a = \left(\frac{m_2}{m_1 + m_2}\right) g$

b. If $m_1 = 5.00\,kg$ and $m_2 = 9.00\,kg$, calculate its acceleration?





$\vec{N} = m\vec{g}$

• For mass m_1: $\Sigma\vec{F} = m\vec{a}$

$\vec{ox}$: $\vec{T} + \vec{f}_r = m_1\vec{a}$

$T - f_r = m_1 a$ (1)

$\vec{oy}$: $\vec{n} + m_1\vec{g} = \vec{0}$

$n - m_1 g = 0$

$n = m_1 g$ (2)

• For mass m_2 $\Sigma\vec{F} = m\vec{a}$

$(\vec{oy})$: $m_2\vec{g} + \vec{T} = m_2\vec{a}$

$m_2 g - T = m_2$ (3)

From eq(1) & (3)

$$+ \begin{cases} \cancel{T} - f_r = m_1 a & (1) \\ m_2 g - \cancel{T} = m_2 a & (2) \end{cases}$$

$m_2 g - f_r = m_1 a + m_2 a$

$m_2 g - \cancel{f_r} = (m_1 + m_2) a$

Assume that, $f_r = 0$: $m_2 g = (m_1 + m_2) a$

$$\Rightarrow \quad a = \frac{m_2 g}{m_1 + m_2} = \left(\frac{m_2}{m_1 + m_2}\right) g$$